

Prolog Program to Implement Pattern Matching

1. Aim

To write and execute a Prolog program to demonstrate pattern matching (unification) between compound terms using facts, rules, and queries.

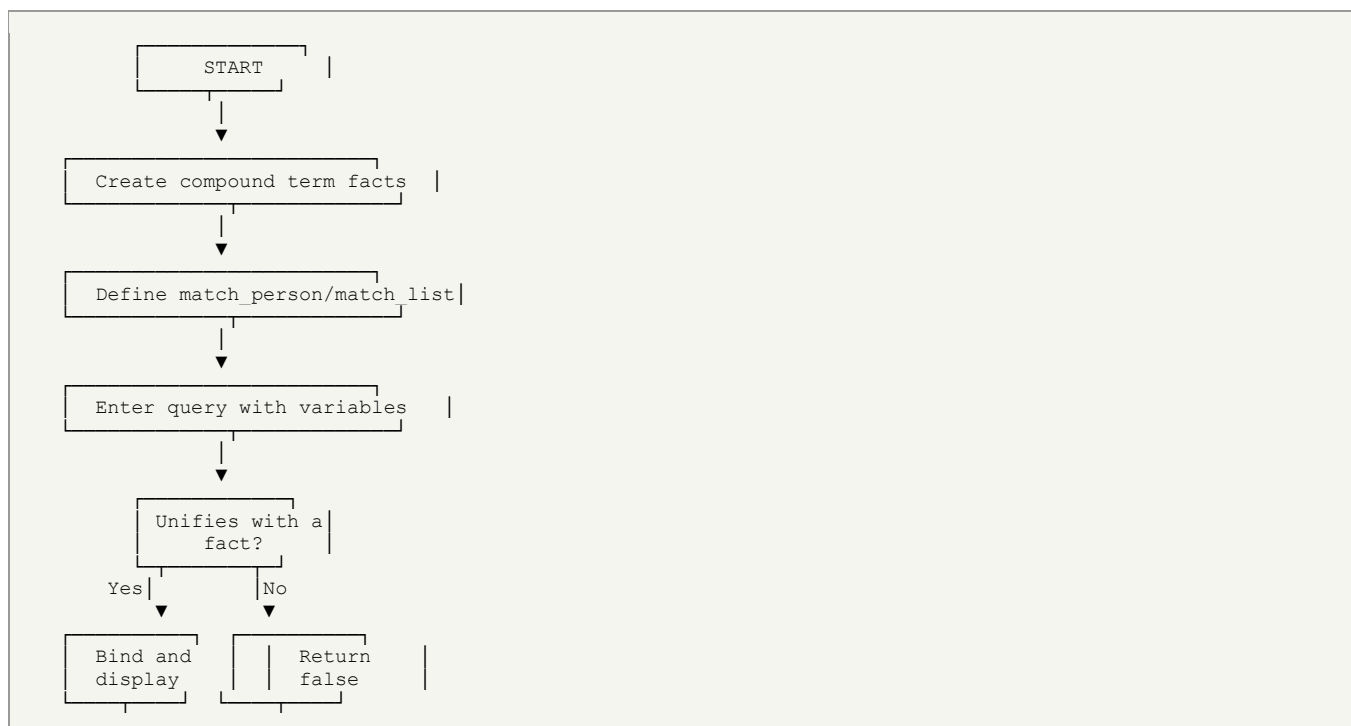
2. Objective

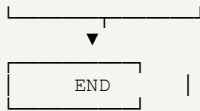
- To understand Prolog's core mechanism of unification.
- To represent structured data using compound terms.
- To match a query term against a fact and extract bound variables.
- To demonstrate pattern matching over lists using recursion.

3. Algorithm

1. Start the program.
2. Create a Prolog database of compound facts, such as person names with structured details.
3. Define a rule `match_person/3` that unifies a query term with a stored compound term.
4. Also define a recursive predicate `match_list/2` to pattern-match corresponding elements of two lists.
5. Enter a query containing unbound variables in place of the values to extract.
6. Prolog attempts to unify the query term with a matching fact.
7. If unification succeeds, the variables are bound and displayed; otherwise Prolog returns false.
8. Stop the program.

4. Flowchart





5. Code with Output

Code:

```
% -----  
% Pattern matching over compound terms  
% -----  
person(john, name(john, smith), age(25)).  
person(mary, name(mary, jones), age(30)).  
  
match_person(Id, Name, Age) :-  
    person(Id, name(Id, Name), age(Age)).  
  
% -----  
% Pattern matching over lists (recursive)  
% -----  
match_list([], []).  
match_list([H|T1], [H|T2]) :-  
    match_list(T1, T2).
```

Output:

```
?- match_person(john, Surname, Age).  
Surname = smith,  
Age = 25.  
  
?- X = point(1, 2), X = point(A, B).  
X = point(1, 2),  
A = 1,  
B = 2.  
  
?- match_list([1,2,3], [1,2,3]).  
true.  
  
?- match_list([1,2,3], [1,2,4]).  
false.
```

6. Result

The Prolog program was successfully implemented to demonstrate pattern matching (unification) over compound terms and lists, correctly binding variables on a match and failing otherwise.

7. Conclusion

Thus, pattern matching was successfully demonstrated using Prolog. The experiment shows how unification underlies both fact retrieval and structural matching between terms of the same shape.